



Chapter 2 Hydrostatics

Section 1 Hydrostatic pressure characteristics

Section 2 Stress equilibrium

Section 3 Pressure profile of a static fluid

Section 4 Manometer

Section 5 Hydrostatic pressure on a plane

Section 6 Hydrostatic pressure on a surface

Section 7 Buoyancy and floating-body stability

Chapter summary





Chapter 2 Hydrostatics

Section 7 Buoyancy and floating-body stability

- 1, Archimedes law, buoyancy, three states of a submerged body
- 2, Balance and stability of submerged body
- 3, Balance and stability of floating body



Section 7 Buoyancy and floating-body stability

1. Archimedes' law, buoyancy, the three states of an immersed object

- Archimedes' law

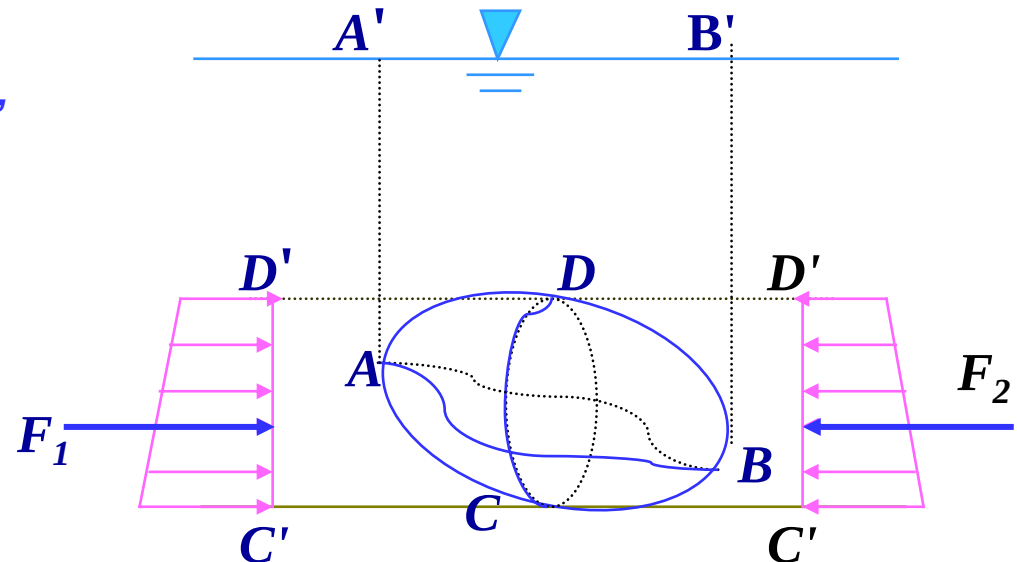
Archimedes' Law: The total hydrostatic force on an object in a static fluid has only a vertical component, and its magnitude is exactly equal to the weight of the liquid displaced by the submerged body.

$$F_x = F_1 - F_2 = 0$$

$$F_z = \rho g V_{A'ACBB'} - \rho g V_{A'ADBB'}$$

$$= \rho g V_{ACBD}$$

= the weight of the liquid displaced by the submerged body (direction up)





- Buoyancy

Buoyancy: the resultant force on an object that has the tendency to push an object toward the surface of the liquid is called buoyancy. The direction of buoyancy is always vertical, upward.

Center of buoyancy: the point of action of buoyancy, which coincides with the centroid of the volume of liquid displaced.

- Three states of a submerged object

When an object immersed in liquid is not supported by other objects, it is affected by gravity **G** and buoyancy **F_z** , so it has the following three states:

(1) sinking body: when **$G > F_z$** , the object sinking to the bottom.

(2) Submerged body: when **$G = F_z$** , it is an object that is submerged in any position in the liquid and kept in balance, that is, suspended.

(3) Floating body: when **$G < F_z$** , it floats up to the water surface and is in a floating state.





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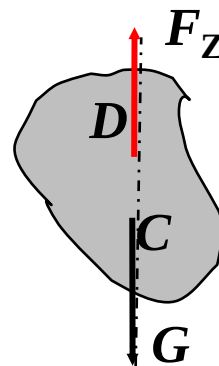
Section 7 Buoyancy and floating-body stability

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2, Balance and stability of submerged body

- A submerged body is in equilibrium when the gravity $\mathbf{G}$ and buoyancy force $\mathbf{F}_z$ are equal in magnitude, opposite in direction, and acting on the same vertical line.
- A submerged body is stable if it returns to its original equilibrium state after encountering (sufficiently small) external disturbances.
- Stability condition: gravity $\mathbf{G}$ is equal to buoyancy $\mathbf{F}_z$, and the center of gravity $\mathbf{C}$ is below the center of buoyancy $\mathbf{D}$.

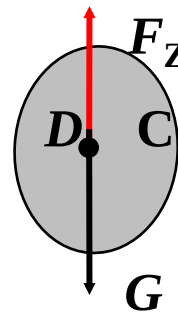
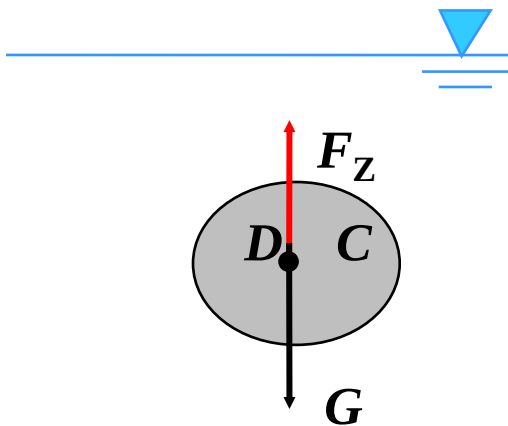


- Three situations of submerged body balance

Neutral: $C = D$

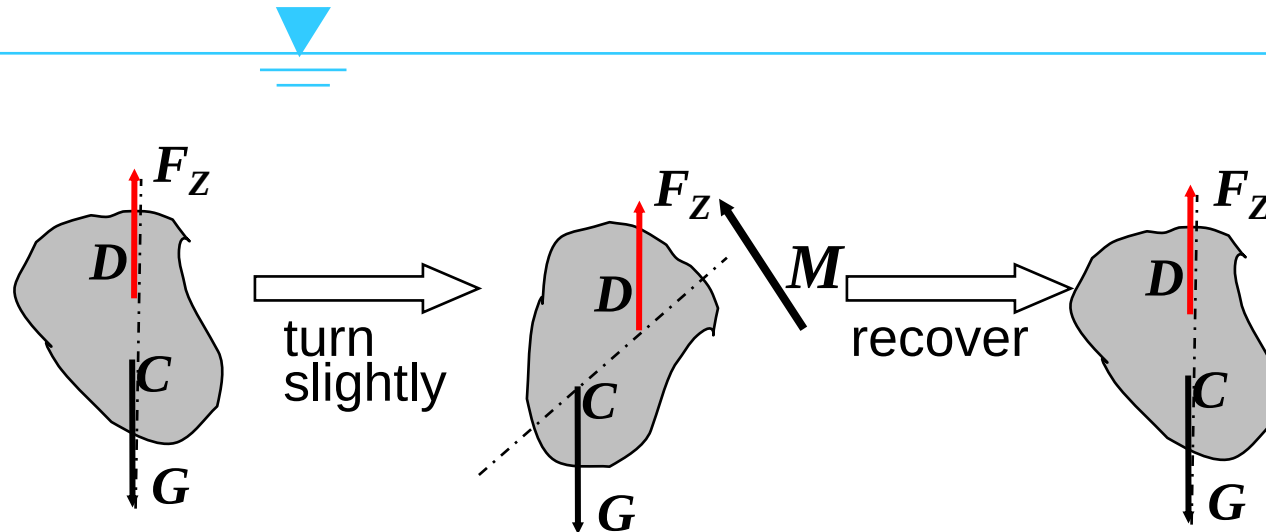
Stable: C below D

Unstable: C above D

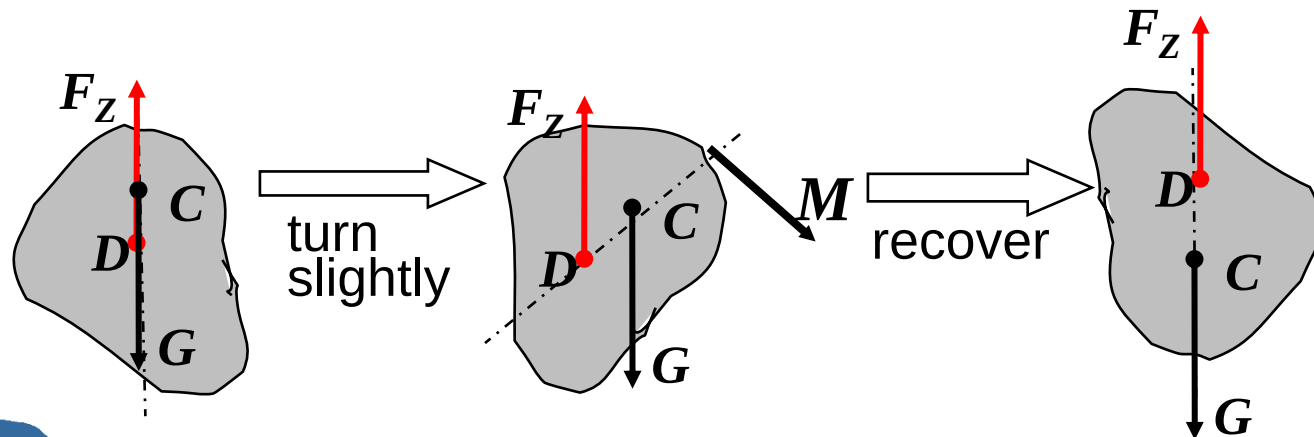


Three situations of submerged body balance

Stable



Unstable



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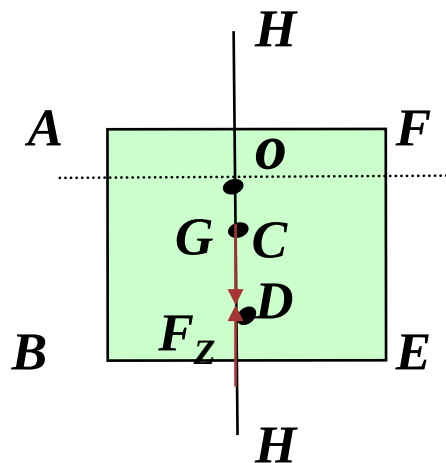
👉 3, Balance and stability of floating body



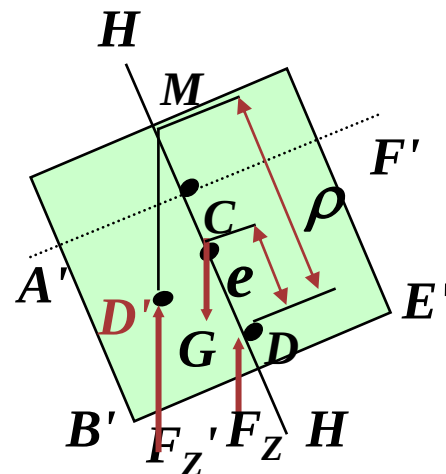
3, Balance and stability of floating body

- A floating body is in equilibrium when the gravity $\mathbf{G}$ and buoyancy force $\mathbf{F}_Z$ are equal in magnitude, opposite in direction, and acting on the same vertical line.

When the center of gravity $\mathbf{C}$ is above that of buoyancy $\mathbf{D}$, a floating body may still be stable, because after tilting, the shape of the part immersed in the water changes, and the position of the center of buoyancy also moves.



(a) Before being disturbed



(b) After being disturbed

Metacenter: Intersection $\mathbf{M}$ between line through $\mathbf{CD}$ and vertical line through $\mathbf{D}'$

Inclination radius ρ : distance from $\mathbf{M}$ to $\mathbf{D}$

Eccentricity e : distance between C and D.

Tilt height: Distance between M and C, $\overline{MC} = \rho - e$

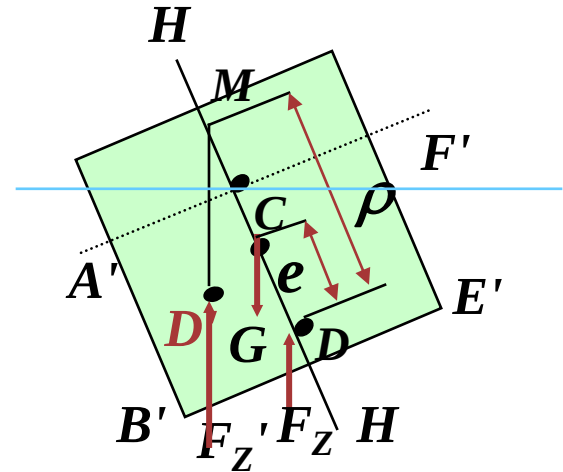
Stability conditions:

$$\left\{ \begin{array}{ll} \text{Stable: } \rho > e & \overline{MC} > 0 \\ \text{Unstable: } \rho < e & \overline{MC} < 0 \\ \text{Neural: } \rho = e & \overline{MC} = 0 \end{array} \right.$$

- Calculation of ρ

For small inclination ($\alpha < 10^\circ$):

$$\rho = I_0/V$$



I_0 — Moment of inertia around its central longitudinal axis;

V — Displaced fluid volume





1. What are the equilibrium and stability conditions of a floating body? When $\rho < \rho_f$ and $\rho = \rho_f$, what is the state of the floating body?

Center of gravity is below tilt height; unstable, neutral.

2. What are the equilibrium and stability conditions of the submerged body? What kind of stability balances does it have?

Gravity G and buoyancy F_z are equal in magnitude, and the center of gravity C is below the center of buoyancy D .

Neutral: $C = D$

Stable: C below D

Unstable: C above D





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Hydrostatics refers to the absence of relative fluid motion. Its core problem is the determination of the equilibrium pressure distribution in still water, and the total hydrostatic force acting on plane and curved surfaces.

1. Two characteristics of static pressure:

- a. It is a compressive stress, directed negative to the normal of a surface.
- b. It is equal in all directions, independent of the orientation of the surface.





2. Pressure representation:

a. Pressure can be divided into absolute pressure, relative pressure, and the vacuum value.

b. Special units of pressure: liquid column height (**mH₂O**) and atmospheric pressure (**at** or **atm**)

3. The concept of isobaric surface: The mass force is vertical to an isobaric surface. If the only mass force is gravity, a horizontal surface is isobaric. At the same horizontal level in a connected, homogeneous fluid under only gravity, the pressure is constant.



4, Stress equilibrium

$$\begin{cases} f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} = 0 \\ f_y - \frac{1}{\rho} \frac{\partial p}{\partial y} = 0 \\ f_z - \frac{1}{\rho} \frac{\partial p}{\partial z} = 0 \end{cases} \quad \text{或} \quad dp = \rho(f_x dx + f_y dy + f_z dz)$$

5, Pressure profile of a static fluid

(1) Basic equations (only gravity): $z + \frac{p}{\rho g} = C; p = p_0 + \rho gh$

(2) Under horizontal acceleration: $p = -\rho(ax + gz)$





6, Total hydrostatic force on a plane

(1) Analytical method:
$$\begin{cases} F = F_c A = \rho g h_c A \\ y_D = y_c + \frac{I_{cx}}{y_c A} \end{cases}$$

(2) Graphical method: $\begin{cases} \mathbf{F} = \text{Volume of pressure diagram} \\ \mathbf{y_D}: \text{Geometric center of pressure diagram} \end{cases}$

7, Total hydrostatic force on a curves surface

$$\begin{cases} F_x = \rho g h_c A \\ F_z = \rho g V_p \\ F = \sqrt{F_x^2 + F_z^2} \\ \alpha = \tan^{-1} \frac{F_z}{F_x} \end{cases}$$

Force on the projected vertical plane

V_p ——Volume of pressure body, composed of:

- (1) The compressed surface itself;
- (2) The vertical planes made of the edges around the curved surface;
- (3) Free surface or extension of free surface.





8, Balance and stability of submerged and floating bodies

Submerged body:

Neutral: $C = D$

Stable: C below D

Unstable: C above D

Floating body:

$$\left\{ \begin{array}{ll} \text{Stable: } \rho > e & \overline{MC} > 0 \\ \text{Unstable: } \rho < e & \overline{MC} < 0 \\ \text{Neural: } \rho = e & \overline{MC} = 0 \end{array} \right.$$

